service claims on the average 8.9 minutes to reach its destination in emergency calls.

Exercise 8.2

1. Mention two branches of statistical inference?
2. What is an estimator?
3. What is an estimate?
4. What is point estimation?
5. What is interval estimation?
6. What is confidence interval?
7. What is null hypothesis? Give an example.
8. Define alternative hypothesis.
9. Define critical region.
10. Define critical value.
11. Define level of significance.
12. What is type I error.
13. What is single tailed test.
14. A sample of 100 items, draw from a universe with mean value 4 and S.D 3, has a mean value 63.5. Is the difference in the mean significant at 0.05 level of significance?
15. A sample of 400 individuals is found to have a mean height of 67.47 inches. Can it be reasonably regarded as a sample from a large population with mean height of 67.39 inches and standard deviation 1.30 inches at 0.05 level of significance?
16. The average score on a nationally administered aptitude test was 76 and the corresponding standard deviation was 8. In order to evaluate a state's education system, the scores of 100 of the state's students were randomly selected. These students had an average score of 72. Test at a significance level of 0.05 if there is a significant difference between the state scores and the national scores.
17. The mean breaking strength of cables supplied by a manufacturer is 1,800 with a standard deviation 100. By a new technique in the manufacturing process it is claimed that the breaking strength of the cables has increased. In order to test this claim a sample of 50 cables is tested. It is found that the mean breaking strength is 1,850. Can you support the claim at 0.01 level of significance.

Exercise 8.3

Choose the correct Answer

1. A ……. may be finite or infinite according as the number of observations or items in it is finite or infinite.
   (a) Population (b) census
   (c) parameter (d) none of these
2. A …… of statistical individuals in a population is called a sample.
   (a) Infinite set (b) finite subset
   (c) finite set (d) entire set
3. A finite subset of statistical individuals in a population is called ……...
   (a) a sample (b) a population
   (c) universe (d) census
4. Any statistical measure computed from sample data is known as ……...
   (a) parameter (b) statistic
   (c) infinite measure (d) uncountable measure
5. A……..is one where each item in the universe has an equal chance of known opportunity of being selected.
   (a) Parameter (b) random sample
   (c) statistic (d) entire data
6. A random sample is a sample selected in such a way that every item in the population has an equal chance of being included
   (a) Harper (b) Fisher
   (c) Karl Pearson (d) Dr. Yates

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